Numbers and Sets

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1. Proofs

A proof is a sequence of true statements without logical gaps, to establish a conclusion.

We have to start somewhere and have agreed assumptions i.e. 'axioms'.

Why do we even prove?
1. We want to know for certain a statement is true.
2. We want to gain insight into why they are true.
3. We might be lucky and find a 'beautiful' proof.

Example (Statements)

1. There are infinitely many primes p s.t. $2p+1$ is also prime.
2. There are infinitely many primes p s.t. one of $p+2, p+4, \ldots, p+240$ is prime.
3. For any $n \in \mathbb{Z}^+$ there is a prime between n and $2n$
4. There is no computer algorithm which will factor an n-digit number in less than n^3 steps.
5. Every non-constant polynomial with complex coefficients has a root in $\mathbb{C}$
6. $\forall m, n \in \mathbb{Z}\ mn = nm$
7. $1 + 1 = 2$

◊

Remark

1. No-one knows if it is true
2. Was proved in Aug 2026
3. Not obvious but true
4. Would be a disaster if true
5. The fundamental theorem of algebra
6. Worth thinking about
7. Does it need proving?

◊

1.1. Some proofs and non-proofs

Proposition

$\forall n \in \mathbb{Z}^+,\ 3 \mid n^3 - n$

~~*Proof.*~~ Observe that $\forall n \in \mathbb{Z}^+, n^3 - n = n(n^2 - 1) = (n+1)n(n-1)$.

Now observe that one of $n-1$, n, $n+1$ must be a multiple of 3. Hence $n^3 - n$ must be a multiple of 3. □

Now for a non-proof

Proposition 1.1.1

$\forall n \in \mathbb{Z}^+$ if n^2 is even then so is n

Proof. Given a positive integer $n \in \mathbb{Z}^+$ then $\exists k \in \mathbb{Z}^+ \ s.t.\ n = 2k$

So $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ and so n^2 is even. □

Clearly this is not a proof of Proposition 1.1.1 as clearly we wanted to prove 'if A then B' but instead we proved 'if B then A' and so Proposition 1.1.1 has not been proved.

Proposition 1.1.2

$\forall n \in \mathbb{Z}^+$ if n^2 is a multiple of 9 then so is n

To show Proposition 1.1.2 is false it is enough to show that 1 value does not satisfy the proposition. i.e. $n = 3$ is a counterexample to the proposition and so Proposition 1.1.2 is false

Now back to

Proposition 1.1.1

$\forall n \in \mathbb{Z}^+$ if n^2 is even then so is n

Proof. Let's suppose the contrary to Proposition 1.1.1

i.e. $\exists n \in \mathbb{Z}^2 \ s.t.\ n$ is odd and n^2 is even. So $n = 2k - 1$ for some $k \in \mathbb{Z}^+$. So $n^2 = (2k-1)^2 = 4k^2 - 4k + 1 = 2(2k^2 - 2k) + 1$, so n^2 is odd.

Therefore contradiction and so if n^2 is even then n is even □

> **Remark**
>
> Showing $A \Rightarrow B$ is equivelant to showing $\neg B \Rightarrow \neg A$

Proposition 1.1.3

The solution to $x^2 - 5x + 6 = 0$ $(*)$ is $x = 2$ or $x = 3$

This is in fact 2 statements in one.
(i) $x = 2$ and $x = 3$ are solutions to $(*)$.
(ii) There are no other solutions to $(*)$

Proof.
(i) If $x = 2$ or $x = 3$, then $x - 2 = 0$ or $x - 3 = 0$, so $(x - 2)(x - 3) = 0$, so $x^2 - 5x + 6 = 0$
(ii) If $x^2 - 5x + 6 = 0$, then $(x - 2)(x - 3) = 0$, then $x - 2 = 0$ or $x - 3 = 0$ so $x = 2$ or $x = 3$ □

Alternatively we could have just written:

Proof.

$x = 2$ or $x = 3$

$\Leftrightarrow x - 2 = 0$ or $x - 3 = 0$

$\Leftrightarrow (x - 2)(x - 3) = 0$

$\Leftrightarrow x^2 - 5x + 6 = 0$

□

> **Remark**
>
> However it is vital that every step that is linked with $\Leftrightarrow$ is actually if and only if.

Proposition 1.1.4

Every positive real is ≥ 1

~~*Proof.*~~

Let r be the least positive real.

Then either $r = 1, r > 1, r < 1$:
- If $r < 1$ then $0 < r^2 < r$, contradiction (as r is not the least positive real).
- If $r > 1$ then $0 < \sqrt{r} < r$, contradiction (as r is not the least positive real).

So as one of these must be true, $r = 1$ □

> **Remark**
>
> Nonsense! Is there a smallest real number r?

> **Warning**
>
> Every claim must be justified.

1.2. Combining Claims

If A and B are assertions/propositions then we can write $A \wedge B$ to mean "A and B", and $A \vee B$ to mean "A or B", and $\neg A$ for "NOT A" The truths of these assertions depend oon the truths of A and B. As summarised below:

A	B	$A \wedge B$	$A \vee B$	$\neg A$	$A \Rightarrow B$
0	0	0	0	1	1
0	1	0	1	1	1
1	0	0	1	0	0
1	1	1	1	0	1

> **Remark**
>
> Note that $\neg(A \wedge B)$ is equivelant to $(\neg A \vee \neg B)$
>
> Also $A \Rightarrow B$ is equivelant to $\neg A \vee B$ and hence $B \vee \neg A$ and hence $\neg B \Rightarrow \neg A$

1.2.1. Negation of qualifiers.

Let x be some object, and $A(x)$ be some property

$\neg(\forall x A(x))$ means $\exists x\ \neg A(x)$

And

$\neg(\exists x A(x))$ means $\forall x\ \neg A(x)$

2. Elementary Number Theory

2.1. The natural numbers

Intuitively the naturals consist of: $1,\ 1+1,\ 1+1+1,\ \ldots$

But how do you know you have captured all the natural numbers? How do you know they are all distinct?

We should assume:

The natural numbers, written $\mathbb{N}$ is a set containing a special element '1' and with a special operation "+1", satisfying:

(i) $\forall n \in \mathbb{N},\ n+1 \neq 1$
(ii) $\forall m, n \in \mathbb{N}$ if $n \neq m$ then $n+1 \neq m+1$
(iii) For some property $P(n)$ if $P(1)$ is true, and $\forall n \in \mathbb{N} P(n) = P(n+1)$, then $\forall n \in \mathbb{N} P(n)$ is true

> **Remark**
>
> - (i)-(iii) are known as the Peano axioms
> - (i) and (ii) capture the distinctness of the natural numbers.
> - (iii) Captures our idea of the Natural numbers being complete. (by taking $P(n) =$ "n is on the list").

Now we can write 2 for $1+1$, 3 for $1+1+1$, and so on.

Definition

The operation "$+k$ " is defined as follows

$n + (k + 1) := (n + k) + 1$

(Then taking $P(n) =$ "$+n$ "is defined, by (iii) is defined.)

Multiplication and Powers are similarly defined.

Now observe the following properties that follow (the intuitive arithmetic properties)

Property 2.1.1

$\forall a, b \in \mathbb{N} a + b = b + a$ (+ is commutative)

Property 2.1.2

$\forall a, b \in \mathbb{N} ab = ba$ (+ is associative)

Property 2.1.3

$\forall a, b, c \in \mathbb{N} (a + b) + c = a + (b + c)$ (+ is associative)

Property 2.1.4

$\forall a, b \in \mathbb{N} (ab)c = a(bc)$ ($\times$ is commutative)

Property 2.1.5

$\forall a, b, c \in \mathbb{N} a(b + c) = ab + ac)$ ($\times$ is distributive)

Definition (Less than)

$a < b$ if and only if $a + c = b$ for some $c \in \mathbb{N}$

And now the Definition (Less than) above gives us some more propeties:

Property 2.1.6

$\forall a, b, c \in \mathbb{N}\ a < b \Rightarrow a + c < b + c$

Property 2.1.7

$\forall a, b, c \in \mathbb{N}\ a < b \Rightarrow ac < bc$

Property 2.1.8

$\forall a, b, c \in \mathbb{N}\ a < b$ and $b < c \Rightarrow a < c$

Property 2.1.9

$\forall a \in \mathbb{N},\ \neg(a < a)$

♠

2.2. Weak induction and Strong Induction

Axiom (iii) is also known as the Weak Principle of Induction (WPI).

Another form is the Strong Principle of Induction (SPI) defined by:

Definition (Strong principle of induction)

(i) $P(1)$ holds
(ii) $\forall n \in \mathbb{N}$ we have $P(m)$ is true $\forall m \leq n \Rightarrow P(n+1)$

♣

Proposition 2.2.1

SPI is equivelant to WPI

♠

Proof. Clearly SPI $\Rightarrow$ WPI

To go the other way (WPI $\Rightarrow$ SPI) apply $Q(n) =$ "P(n) holds $\forall m \leq n$ " □

Definition (Well ordering principle (WOP))

if $P(n)$ holds for some $n \in \mathbb{N}$ then there is a least $n \in \mathbb{N} s.t.\ P(n)$ holds.

♣

Theorem 2.2.2

SPI⇒WOP

♡

Proof. Suppose the contrary i.e. there is no least $n \in \mathbb{N} s.t.\ P(n)$ holds. Now we want to show that $P(n)$ does not hold $\forall n \in \mathbb{N}$ using SPI

Consider $Q(n) = \neg P(n)$

Certainly $P(1)$ false (otherwise would be least value to satisfy $P(n)$), and so $Q(1)$ is true.

Given $n \in \mathbb{N}$, suppose $Q(k)$ is true $\forall k < n$. Then $P(k)$ is false. So $P(n)$ is false otherwise n would be the minimal element, for which P holds, hence $Q(n)$ is true.

So both (i) and (ii) of Definition (Strong principle of induction) are satisfied and so $Q(n)$ is true for all $n \in \mathbb{N}$. Thus $P(n)$ is false $\forall n \in \mathbb{N}$ contradicting Definition (Well ordering principle (WOP))

□

WOP allows us to prove $P(n)$ is true by using the minimal counterexample to form a contradiction.

2.3. The integers

The integers, written $\mathbb{Z}$ consist of all symbols $n, -n$, where $n \in \mathbb{N}$ along with 0. You can define $+$ and $\times$ on $\mathbb{Z}$ from the definitions for $\mathbb{N}$ and you can also check that rules (1)-(5) also still hold.

Property 2.3.1

$\forall a \in \mathbb{Z},\ a + 0 = a$ (identity for $+$)

Property 2.3.2

$\forall a \in \mathbb{Z} \exists b \in \mathbb{Z}\ s.t.\ a + b = 0$ (inverses for addition)

Definition (Less than on integers)

for $a, b \in \mathbb{Z}$

$a < b$ if $a + c = b$ for some $c \in \mathbb{N}$

It then follows that rules Property 2.1.6,Property 2.1.8, and Property 2.1.9 still hold but Property 2.1.7 must be modified.

Property 2.3.3

$\forall a, b, c \in \mathbb{Z}$, if $a < b \wedge c > 0 \Rightarrow ac < bc$

2.4. The rationals

The rationals written $\mathbb{Q}$, consist of all expressions $\frac{a}{b}$ where $a, b \in \mathbb{Z},\ b \neq 0$.

Also $\frac{a}{b} = \frac{c}{d} \iff ad = bc$

Definition (Rational operators)

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

It is then easy enough to show all the above rules of arithmetic apply.

Remark

$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$

2.5. Prime Numbers

Definition (Divides)

Given $a, b \in \mathbb{Z}$ we say 'a divides b', 'a is a factor of b' or $a \mid b$ if and only if

$\exists\ c \in \mathbb{Z}\ s.t.,\ ac = b$

Remark

$\forall\ b \in \mathbb{Z},\ \pm 1$ and $\pm b$ are always factors of b. All factors are called 'proper' or 'non-trivial' factors of b

Definition (Prime number)

$n \in \mathbb{N}$ is called a prime number if and only if $n \geq 2$ and its only factors are the trivial ones (± 1 and $\pm n$)

Definition (Composite Number)

$n \in \mathbb{N}$ is called a composite number if and only if $n \geq 2$ and it is not prime.

Proposition 2.5.1

Every natural number can be written as the product of primes.

Proof. By induction on $n \in \mathbb{N}, n \geq 2$:

Base Case $n = 2$: 2 is prime so can be written as a product of primes.

Inductive Step: Assume the statement holds for all numbers up to and including $n - 1$.

Then if n is prime then done. If n is composite then $n = ab$ for some $a, b \in \mathbb{N}, a, b \geq 2$. By the hypothesis (and as $a, b < n$) $a = p_1, p_2, ..., p_n$ and $b = q_1, q_2, ..., q_m$ for some primes $p_1, p_2, ..., p_n, q_1, ..., q_m$ and $n, m \in \mathbb{N}$. So $n = p_1 p_2 p_3 ... p_n q_1 q_2 ... q_m$ and so can be expressed as a product of primes. □

Theorem 2.5.2

There are infinitely many prime numbers.

Proof. Suppose for sake of contradiction that there are finitely many primes. Then all the primes are $p_1, p_2, ..., p_k$ for some $k \in \mathbb{N}$

Now take $N = \prod_{i=1}^{k} p_i + 1$. Now $p_1 \nmid N$ as otherwise $p_1 \mid 1$ but $p_1 \geq 2$. Likewise $p_2, ..., p_k$ don't divide N. So N can't be written as a product of primes contradicting Proposition 2.5.1 □

Note

Our proof of Proposition 2.5.1 does not guarantee uniqueness.

However $21 = 3 \times 7$ is clearly unique.

How about $295869 = 3 \times 7 \times 73 \times 193$

Why does $9040 \times 40099 \neq 6701 \times 54151$

Proposition 2.5.3

If p is prime and $p \mid ab$ then $p \mid a$ or $p \mid b$.

To prove Proposition 2.5.3 we need more tools.

Definition 2.5.4 (Highest Common Factor (HCF))

Given $a, b \in \mathbb{N}$ and $c \in \mathbb{N}$ is,

the highest common factor (HCF) or greatest common divisor (GCD) of a and b if:

(i) $c \mid a$ and $c \mid b$ (c is a common divisor of a, b)
(ii) if for some $d \in \mathbb{N}$, $d \mid a$ and $d \mid b \Rightarrow d \mid c$

Notation

We also write $c = hcf(a, b) = gcd(a, b) = (a, b)$

Example

The factors of 12 are $1, 2, 3, 4, 6, 12$.

The factors of 18 are $1, 2, 3, 6, 18$

So the common factors are $1, 2, 3, 6$.

Now to use hcf we need to show it always exists.

Proposition 2.5.5 (Division Algorithm)

let $n, k \in \mathbb{N}$, then we can write $n = qk + r$, where $q, r \in \mathbb{Z}$, with $0 \leq r \leq k - 1$.

r can be referred to as the remainder, q can be referred to as the quotient.

Proof. By induction on $n \in \mathbb{N}$.

Base case: true for $n = 1$ ($q = 0, r = 1$)

Inductive step: suppose $n \geq 2$ and the statement holds for $n - 1$, i.e. $n - 1 = qk + r$ for some $q, r \in \mathbb{Z}, 0 \leq r \leq k - 1$.

Now if $r < k - 1$ then $n = (n - 1) + 1 = qk + r + 1 = qk + (r + 1)$. With $q, r \in \mathbb{Z}$ and $0 \leq r \leq k - 1$.

Now if $r = k - 1$ then $n = qk + k - 1 + 1 = (q + 1)k + 0$. So also holds. □

Corollary 2.5.5.1

The q, r obtained are unique to each number.

Proof. If $n = qk + r = q`k + r`$. Then $(q - q`)k = r` - r$, with $-k < RHS < k$, so $q` = q, r = r`$. □

2.5.1. Euclid's Algorithm

INPUT	a b
STEP 1	$a = q_1 b + r_1, 0 \leq r_1 \leq b - 1$
STEP 2	$b = q_2 r_1 + r_2, 0 \leq r_2 \leq r_1 - 1 \leq b - 2$
STEP 3	$r_1 = q_3 r_2 + r_3, 0 \leq r_3 \leq r_2 - 1 \leq b - 3$
...	...
STEP n	$r_{n-2} = q_n r_{n-1} + r_n, 0 \leq r_n \leq r_{n-1} - 1 \leq b - n$
STEP $n+1$	$r_{n-1} = q_{n+1} r_n + r_{n+1}, \quad r_{n+1} = 0$

Then the output is r_n

Observe that the algorithm will terminate in $n < b$ steps since $b > r_1 > r_2 > ... > r_n > 0$

Theorem 2.5.6

The output of Euclid's Algorithm with input a, b is $hcf(a, b)$

Proof.

(i) Have $r_n \mid r_{n-1}$ as $r_{n+1} = 0$. So $r_n \mid r_{n-2}$ so $r_n \mid r_i, \forall i = 1, 2, ..., n-1$ (inductively).

Hence $r_n \mid b$, and hence $r_n \mid a$

(ii) Given $d \ s.t. d \mid a$ and $d \mid b$, we have $d \mid r_1, d \mid r_2, ..., d \mid r_n$

So $d \mid r_n$

So r_n satisfies Definition 2.5.4 and so $hcf(a, b) = r_n$ □

Example

To obtain $hcf(87, 52)$:

$$87 = 1 \times 52 + 35$$

$$52 = 1 * 35 + 17$$

$$35 = 2 * 17 + 1$$

$$17 = 17 \times 1$$

So $hcf(87, 52) = 1$